VISIONINSPECT AI

Milestone 1 – Project Initialization, System Design and Core Foundation

Milestone	Milestone 1
Focus	Establish the project foundation, architecture, authentication, database and initial full-stack structure.

1. Introduction

VisionInspect AI is an AI-powered manufacturing quality inspection platform designed to support automated visual inspection of manufactured products. The first milestone focused on establishing the complete project foundation required for subsequent AI, analytics and deployment work.

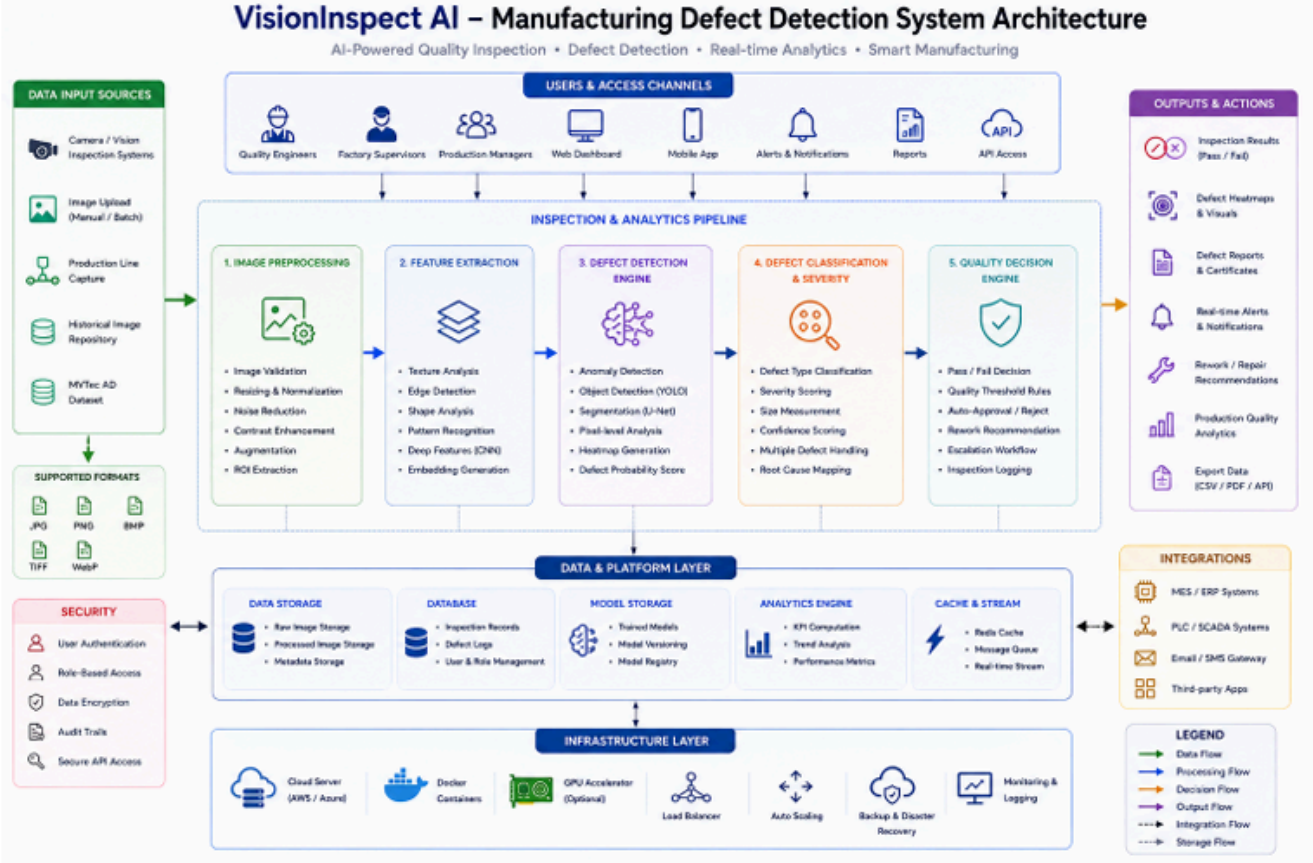
2. Milestone 1 Objectives

- Understand the manufacturing defect detection and quality inspection problem.
- Define the functional modules and overall system architecture.
- Set up the frontend and backend project structure.
- Establish database connectivity and the initial data model.
- Implement user authentication and role-based access foundations.
- Create the initial dashboard and navigation structure.
- Prepare the project repository and development environment for further milestones.

3. System Architecture

The project was organized as a full-stack application consisting of a React frontend, FastAPI backend and PostgreSQL database. The architecture was designed so that computer-vision services could be integrated into the backend without changing the overall application structure.

- Frontend: React-based user interface for login, dashboard and inspection workflows.
- Backend: FastAPI REST API for authentication, inspection processing and analytics services.
- Database: PostgreSQL for users and inspection records.
- AI layer: Computer-vision services integrated into the backend.
- Deployment foundation: Docker-based containerization prepared for cloud deployment.



4. Project Modules Established

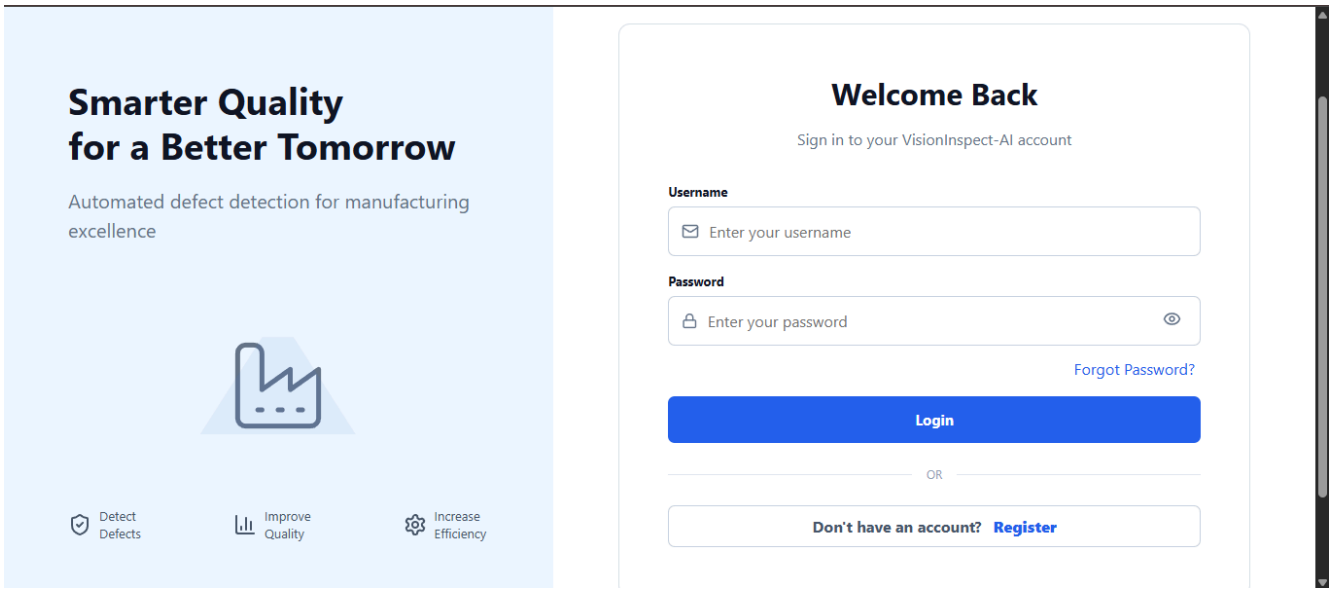
- User Management
- Image Acquisition
- Image Processing
- Defect Detection
- Defect Classification
- Quality Control
- Manufacturing Analytics Dashboard

5. Authentication and Role-Based Access

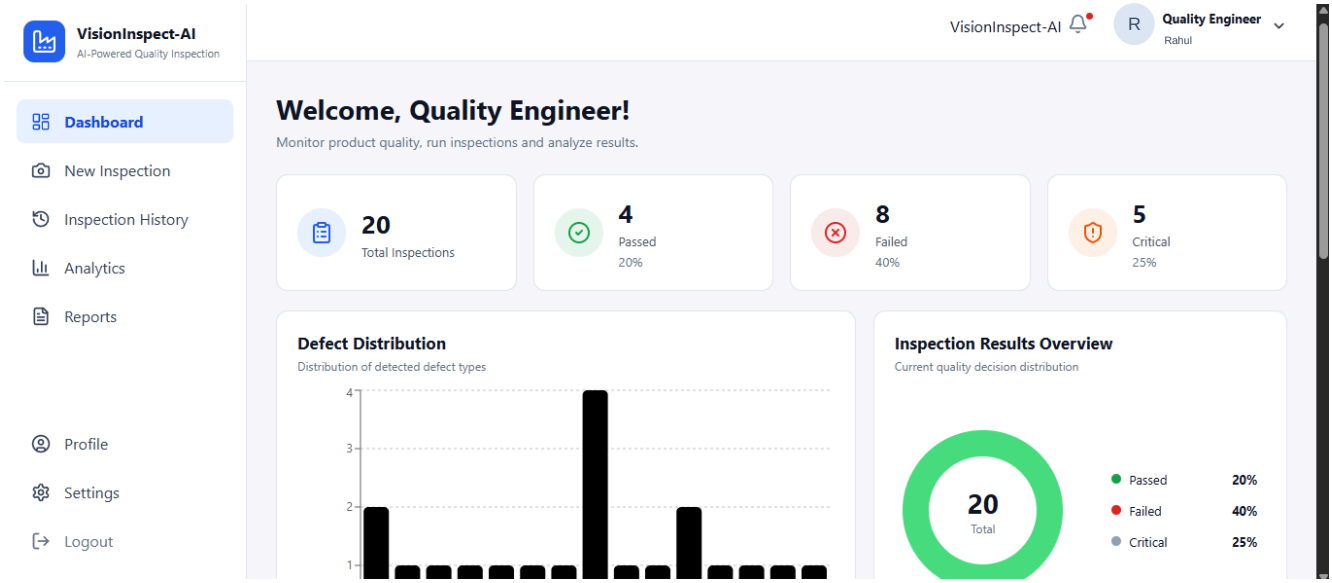
The application includes registration and login functionality with authenticated access to the application. The system supports role-oriented interfaces, including Quality Engineer and Factory Supervisor workflows.

- User registration and login APIs were established.
- Authenticated sessions are used to access protected application functions.
- Quality Engineer functionality includes inspection execution and detailed inspection views.
- Factory Supervisor functionality provides production-oriented monitoring and analytics.

Login / Registration:



Initial Dashboard:



6. Database Foundation

A relational database foundation was established for storing application users and inspection information. The database design supports later analytics, inspection history and reporting features.

- PostgreSQL database configured for the deployed application.
- Users table used for authentication and role information.
- Inspections table used to persist inspection results and metadata.

7. Initial Development Environment

- Git/GitHub repository established for source-code management.

- React frontend and FastAPI backend created as separate application layers.
- Python virtual environment and required development dependencies configured.
- Docker configuration introduced as the basis for reproducible deployment.

8. Milestone 1 Outcome

Milestone 1 established the complete foundation of VisionInspect AI. At the end of this milestone, the project had a defined architecture, working application structure, authentication foundation, database foundation and navigation/dashboard framework. These components provided the base for implementing AI-powered inspection in the following milestones.

9. Conclusion

Milestone 1 successfully completed the project initialization and core application foundation. The system was ready to proceed to image acquisition, preprocessing and AI-based defect detection.